



CODIORA
OUR CODE BUILDS YOUR VISION

WEEK 5 TASKS

1. Frontend Development

Task Title:

Professional Portfolio Deployment & Optimization

Objective

Over the past four weeks, you have built and enhanced a dynamic portfolio website with modern UI components and backend integration. This week, your objective is to optimize your application for production, improve user experience, and deploy your project online. By the end of this task, your portfolio should be fast, responsive, accessible, and publicly available.

Requirements

1. Portfolio Deployment

Deploy your portfolio to a public hosting platform.

Supported platforms include:

- Vercel
- Netlify
- GitHub Pages (for static websites)

Your deployed portfolio should be accessible through a public URL.

2. Performance Optimization

Improve your application's overall performance by:

- Optimizing images
- Removing unused CSS
- Lazy loading images
- Optimizing JavaScript bundles
- Reducing page loading time

3. SEO Improvements

Improve the visibility of your website by implementing:

- Proper Page Title
- Meta Description
- Open Graph Tags
- Favicon
- Semantic HTML

4. Accessibility Improvements

Ensure your website is accessible by implementing:

- Keyboard Navigation
- ARIA Labels
- Image Alt Text
- Proper Heading Structure
- Accessible Color Contrast

5. Code Refactoring

Improve code quality by:

- Organizing Components
- Removing Duplicate Code

- Creating Reusable Components
- Organizing Project Structure

6. Project Documentation

Create a professional README file containing:

- Project Overview
- Features
- Technologies Used
- Installation Steps
- Deployment Link
- Screenshots

Recommended Technologies

- React
- Next.js
- Tailwind CSS
- Vercel
- Netlify
- Lighthouse

Evaluation Criteria

- Deployment
- Performance
- Accessibility
- Code Quality
- Documentation

2. Backend Development

Task Title:

Production Ready Backend API

Objective

Your backend application now contains authentication and portfolio management APIs. This week, you will focus on transforming your backend into a production-ready service by improving security, documentation, performance, and deployment practices.

Requirements

1. API Deployment

Deploy your backend using any cloud hosting platform such as:

- Render
- Railway
- Fly.io

Ensure all API endpoints are publicly accessible.

2. Environment Configuration

Move all sensitive configuration into environment variables, including:

- Database Credentials
- JWT Secret
- API Keys

Never expose sensitive information inside your source code.

3. Advanced Validation

Implement server-side validation for all API endpoints using validation libraries.

Validate:

- Email Format
- Password Strength
- Required Fields
- Data Types

4. Logging System

Implement application logging to record:

- User Login Activity
- API Errors
- Database Errors
- Server Events

5. API Documentation

Prepare complete API documentation including:

- Endpoint Description
- Request Body
- Response Format
- Authentication
- Error Responses

6. Backend Testing

Test all API endpoints using Postman or Swagger and ensure every endpoint returns the correct response.

Recommended Technologies

- Node.js

- Express
- MongoDB
- PostgreSQL
- Swagger
- Postman
- Render

Evaluation Criteria

- Deployment
 - Security
 - Documentation
 - Validation
 - API Reliability
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3. Full Stack Development

Task Title:

Production Ready Portfolio Management System

Objective

Your application is now a fully functional Portfolio Management System. This week, your goal is to optimize, deploy, and document your project while ensuring every module works seamlessly together.

Requirements

1. Deploy Full Stack Project

Deploy:

- Frontend
- Backend
- Database

Ensure the complete application is publicly accessible.

2. Complete Integration Testing

Verify:

- Authentication
- CRUD Operations
- Dashboard
- Image Upload
- Search
- Filters

3. Dashboard Improvements

Add:

- Success Notifications
- Error Notifications
- Loading Indicators
- Better Navigation

4. User Profile Management

Allow users to:

- Change Password
- Update Profile
- Upload Profile Picture

5. Documentation

Create professional documentation including:

- Installation Guide
- Folder Structure
- API Documentation
- Screenshots
- Deployment Links

6. Code Optimization

Improve:

- Component Structure
- API Calls

- Database Queries
- Overall Performance

Recommended Technologies

- MERN Stack
- Next.js
- Laravel
- Django
- Render
- Vercel

Evaluation Criteria

- Deployment
 - Integration
 - Documentation
 - User Experience
 - Project Quality
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4. Mobile App Development

Task Title:

Production Ready Portfolio Mobile Application

Objective

Your portfolio application now includes authentication, project management, and API integration. This week, you will focus on improving performance, testing, deployment readiness, and creating a polished user experience suitable for professional use.

Requirements

1. App Optimization

Improve:

- Screen Loading
- Navigation
- API Calls
- State Management

2. User Experience Improvements

Implement:

- Loading Indicators
- Error Messages
- Empty States
- Success Messages

3. Offline Support

Allow users to view previously loaded data when internet connectivity is unavailable by implementing local caching.

4. Profile Management

Allow users to:

- Update Profile
- Change Profile Image
- Edit Contact Information

5. App Testing

Test every screen for:

- Navigation
- Responsiveness
- API Integration
- Error Handling

6. Project Documentation

Prepare documentation including:

- Features
- Technologies
- Installation
- Screenshots

Recommended Technologies

- Flutter
- React Native
- Firebase
- Hive

- Shared Preferences
- Riverpod

Evaluation Criteria

- Performance
 - User Experience
 - Code Quality
 - Testing
 - Documentation
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5. Graphic Design

Task Title:

Complete Professional Brand Identity System

Objective

You have developed several branding assets over the previous weeks. This week, your goal is to combine those assets into a complete and consistent professional brand identity that can be used across portfolios, freelance platforms, presentations, and social media. Focus on maintaining visual consistency, improving presentation quality, and preparing designs that are ready for real-world use.

Requirements

1. Portfolio Presentation Design

Design a complete portfolio presentation including:

- Cover Page
- About Section
- Services
- Projects
- Contact Information

2. Marketing Materials

Design:

- Flyer
- Brochure
- Social Media Templates
- Business Card

3. UI Design Showcase

Present your UI designs inside realistic device mockups.

4. Brand Guidelines

Create a branding guide containing:

- Logo Usage
- Typography
- Color Palette
- Icons
- Spacing Rules

5. Design Case Study

Explain one complete project by showing:

- Research
- Wireframes
- Final Design
- Mockups

6. Design Presentation

Export all designs into one professional presentation or portfolio document.

Recommended Tools

- Figma
- Photoshop
- Illustrator
- Canva

Evaluation Criteria

- Branding Consistency
 - Creativity
 - Professional Presentation
 - Visual Hierarchy
 - Design Quality
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6. Data Analysis

Task Title:

Business Intelligence Reporting & Data Storytelling

Objective

During the previous weeks, you cleaned data, built dashboards, and analyzed portfolio-related information. This week, your objective is to transform your analysis into a professional business report that communicates meaningful insights. Focus on storytelling with data by presenting clear visualizations, actionable recommendations, and well-structured reports that help stakeholders make informed decisions.

Requirements

1. Dashboard Refinement

Improve your dashboard by:

- Enhancing Visual Design
- Improving Layout
- Organizing Charts
- Adding Interactive Filters

2. Advanced Analysis

Perform:

- Trend Analysis
- Comparative Analysis
- Category Analysis
- Performance Analysis

3. Business Report

Prepare a report containing:

- Executive Summary
- Dashboard Insights
- Business Recommendations
- Future Improvements

4. Data Storytelling

Present your findings using clear charts, visuals, and explanations that guide the reader through the key insights.

5. Dashboard Optimization

Improve:

- Dashboard Performance
- User Navigation
- KPI Organization

6. Presentation

Prepare a professional presentation explaining:

- Dataset
- Analysis Process
- Dashboard
- Insights
- Recommendations

Recommended Technologies

- Power BI
- Tableau
- Python
- SQL
- Excel
- Pandas

Evaluation Criteria

- Data Accuracy
 - Dashboard Design
 - Business Insights
 - Storytelling
 - Presentation Quality
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Week 5 Outcome

By the end of **Week 5**, every intern should have a **production-ready, well-documented, and professionally presented project**.

- **Frontend Developers** will deploy and optimize a high-performance portfolio website.
- **Backend Developers** will publish secure, validated, and documented APIs suitable for production.
- **Full Stack Developers** will deliver a complete Portfolio Management System with deployment, testing, and polished user experience.
- **Mobile App Developers** will refine their applications with offline support, testing, and production-ready optimizations.
- **Graphic Designers** will produce a comprehensive brand identity system accompanied by professional presentations and case studies.
- **Data Analysts** will transform analytical dashboards into business intelligence reports that effectively communicate insights and recommendations.

LinkedIn Submission

After completing your task:

1. Share your work on LinkedIn.
2. Include screenshots of your progress.
3. Briefly explain what you built and what you learned.
4. Submit your LinkedIn post link through the provided **Week 5** Google Form.

Submission Guidelines

- Use any technology of your choice.
- Focus on quality rather than quantity.
- Keep your work organized and professional.
- Creativity and additional features are encouraged but not mandatory.

Submission Deadline

Submit your work before the assigned deadline.

Best of Luck!

Codiora House (Private) Limited

Our Code Builds Your Vision